

Technical Work Documentation

Workbook: NewCo_Campaign_Analysis.xlsx

Primary Analyst: BCG X Data Science Team

Data Engine: Microsoft Excel Pivot Engine

Workbook Architecture

The Excel model is structured across distinct tabs to isolate raw data, analytical aggregations, and visual reporting outputs:

- **Raw Data:** Contains the full dataset.
- **Cleaned data:** Contains the full audited dataset (100% unique Interaction ID rows).
- **Channel Aggregations:** Stores summary tables broken down by Channel.
- **Campaign Aggregations:** Stores summary tables broken down by Campaign.
- **Customer Segmentation:** Holds filtered cross-tabulations isolating New vs. Existing customer groups.
- **Executive_Dashboard:** Visual presentation tab hosting dynamic charts linked to slicers.

Technical Formulas and Logic Applied

- **Total Channel Sales:** = SUM(Raw_Data[Sales (\$)])

- **Total Campaign Sales:** =SUM(Raw_Data[Sales (\$)])
- **New Customer Revenue Attribution:**
=SUMIFS(Raw_Data[Sales (\$)], Raw_Data[Customer Type], "New")
- **Existing Customer Revenue Attribution:**
=SUMIFS(Raw_Data[Sales (\$)], Raw_Data[Customer Type], "Existing")

Pivot Table Output Matrices

Matrix 1: Total Sales by Channel

- **Rows:** Channel
- **Values:** Sum of Sales (\$)

Total Sales By Channel

Channel	Sum of Sales (\$)
Email	\$7,909.31
Instagram	\$3,873.41
Web	
Banner	\$2,876.88
Grand Total	\$14,659.60

Matrix 2: Total Sales by Campaign

- **Rows:** Campaign
- **Values:** Sum of Sales (\$)

Total Sales By Campaign

Campaign Sales	Sum of Sales (\$)
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A	\$5,869.56
B	\$8,790.04

Grand Total	\$14,659.60
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Matrix 3: Full Cross-Tabulation (Channel x Campaign)

- **Filters:** Customer Type = (All)
- **Rows:** Channel
- **Columns:** Campaign
- **Values:** Sum of Sales (\$)

Customer Type (All)

Sum of Sales (\$)	Column Labels			
Row Labels	A	B	Grand Total	
Email	\$ 2,952.37	\$4,956.94	\$ 7,909.31	
Instagram	\$ 2,162.90	\$1,710.51	\$ 3,873.41	
Web Banner	\$ 754.29	\$2,122.59	\$ 2,876.88	
Grand Total	\$ 5,869.56	\$8,790.04	\$14,659.60	

Matrix 4: New Customer Acquisition Matrix

- **Slicer Filter:** Customer Type = New
- **Rows:** Channel
- **Columns:** Campaign
- **Values:** Sum of Sales (\$)

Customer Type New

Sum of Sales (\$) Column Labels

Row Labels	A	B	Grand Total
Email	\$ 2,124.47	\$1,382.38	\$ 3,506.85
Instagram	\$ 969.46	\$ 416.14	\$ 1,385.60
Web Banner	\$ 633.06	\$ 318.92	\$ 951.98
Grand Total	\$ 3,726.99	\$2,117.44	\$ 5,844.43

Matrix 5: Existing Customer Re-engagement Matrix

- **Slicer Filter:** Customer Type = Existing
- **Rows:** Channel
- **Columns:** Campaign
- **Values:** Sum of Sales (\$)

Customer Type Existing

Sum of Sales (\$)	Column Labels			
Row Labels	A	B	Grand Total	
Email	\$ 827.90	\$3,574.56	\$ 4,402.46	
Instagram	\$ 1,193.44	\$1,294.37	\$ 2,487.81	
Web Banner	\$ 121.23	\$1,803.67	\$ 1,924.90	
Grand Total	\$ 2,142.57	\$6,672.60	\$ 8,815.17	

Matrix 6: Channel Conversion Efficiency & Volume Contribution Matrix

- **Filter:** Customer Type = (All)
- **Rows:** Channel
- **Columns:** Campaign Type

- **Values:** Sum of Converted

Customer Type (All)					
Sum of Converted (1=yes, 0=no)	Column Labels			Total Conversion	Volume Share Conversions
Row Labels	A	B	Grand Total		
Email	41	72	113	113	47%
Instagram	35	29	64	64	27%
Web Banner	17	45	62	62	26%
Grand Total	93	146	239	239	100%
Overall Campaign Conversion Rate 48%					

Core Insight Uncovered via Technical Analysis

Applying the Customer Type slicer alongside channel conversion analysis exposes a critical strategic divergence in campaign effectiveness:

- **New Customer Acquisition (Growth Driver):** Campaign A is the primary engine for acquiring new buyers, generating **63.77%** of all **New Customer** revenue (\$3,726.99 out of \$5,844.43). **Email** and **Instagram** served as the most influential entry touchpoints, generating \$2,124.47 and \$969.46 in new customer sales respectively.
- **Existing Customer Retention (Value Driver):** Campaign B dominates existing customer re-engagement, driving **75.69%** of all **Existing Customer** revenue (\$6,672.60 out of \$8,815.17) and capturing **61.09%** of total overall

conversions (146 out of 239). **Email** was the standout channel for repeat purchases, driving \$3,574.56 in existing customer revenue.

- **Channel Conversion Efficiency:** While Campaign B delivered higher raw conversion volume via Email (72 conversions) and Web Banners (45 conversions), **Campaign A achieved higher efficiency on Instagram**, delivering 35 conversions compared to Campaign B's 29. Overall, **Email** proved to be the most efficient channel across both customer segments, delivering a **22.60% channel conversion rate** and accounting for **47.28% of total campaign conversions**.